EXERCÍCIOS DA APOSTILA

$y = \arcsen(x^2)$

$$y' = \frac{1}{\sqrt{1-(x^2)^2}}(x^2)' =$$

$$= \frac{1}{\sqrt{1-x^4}} \cdot 2x = \frac{2x}{\sqrt{1-x^4}}$$

- $y = \arccos(1-x^2)$

$$y' = -\frac{1}{\sqrt{1-(1-x^2)^2}} \cdot (1-x^2)' =$$

$$= -\frac{1}{\sqrt{1-(1-2x^2+x^4)}} \cdot (-2x)$$

$$= \frac{2x}{\sqrt{-x^4+2x^2}} =$$

$$= \frac{2x}{\sqrt{x^2(-x^2+2)}} = \frac{2}{\sqrt{-x^2+2}}$$

ADUEMS

- $y = \operatorname{arc\,sec}(1 - x^2)$

$$y' = \frac{1}{|1-x^2|\sqrt{(1-x^2)^2 - 1}} \cdot (1-x^2)'$$

$$= \frac{-2x}{|1-x^2|\sqrt{(1-2x^2+x^4)-1}} =$$

$$= \frac{-2x}{|1-x^2|\sqrt{x^4-2x^2}} =$$

$$= \frac{-2x}{|1-x^2|\sqrt{x^2(x^2-2)}} =$$

$$= \frac{-2}{|1-x^2|\sqrt{x^2-2}}$$

- $y = \operatorname{arc\,tg}\left(\frac{3}{x}\right)$

$$y' = \frac{1}{1+\left(\frac{3}{x}\right)^2} \cdot \left(3/x\right)'$$

$$= \frac{\frac{-3}{x^2}}{1+\left(\frac{9}{x^2}\right)} = \frac{\frac{-3}{x^2}}{\frac{x^2+9}{x^2}}$$

$$= \frac{-3}{x^2+9}$$